

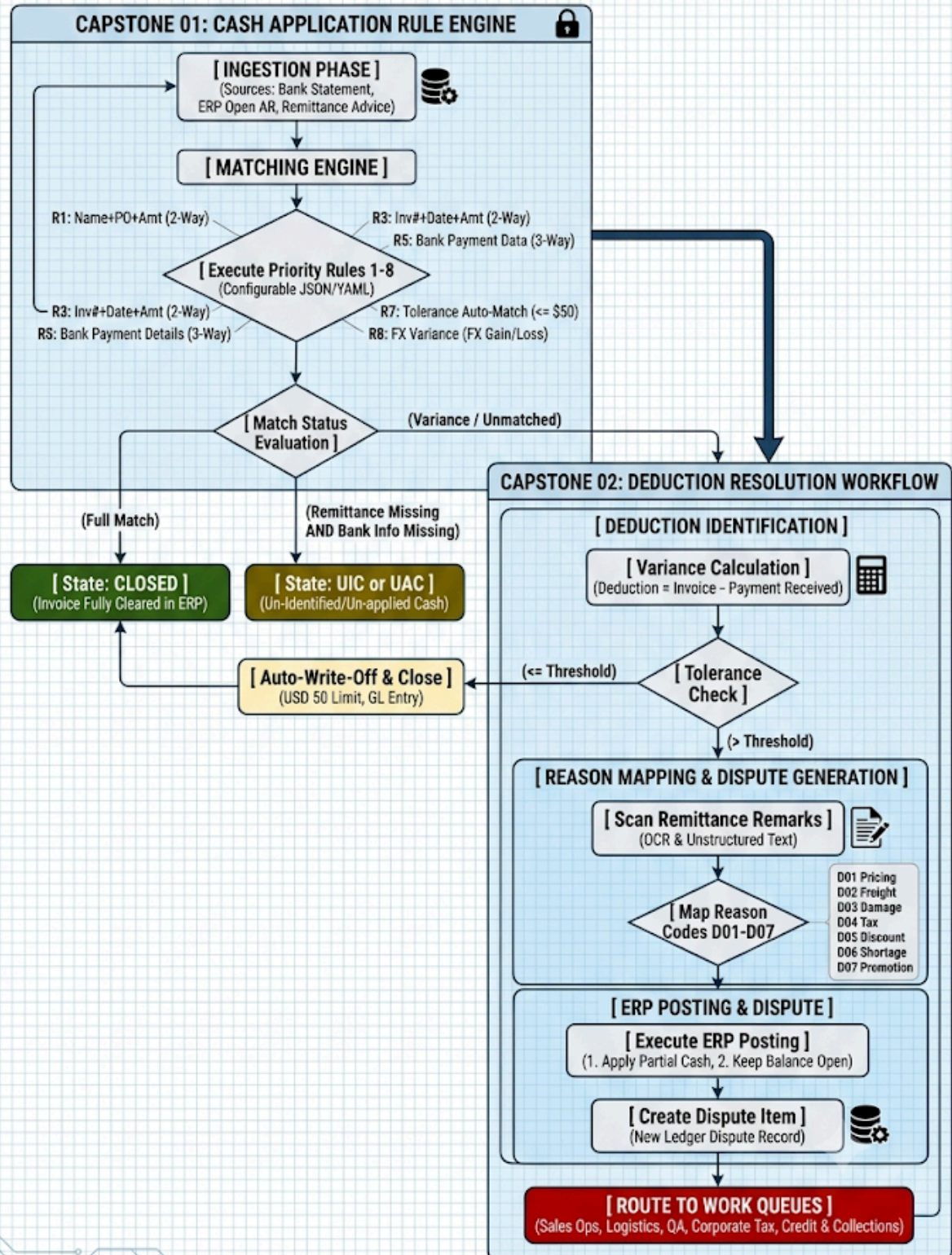
Integrated Capstone Specification: Enterprise Rule Engine & Deduction Resolution Architecture

Technical Overview & Continuum Architecture

The Cash Application and Deduction Management System operates as a two-stage sequential processing engine:

1. **Capstone 01 (Configurable Cash Application Rule Engine):** Handles incoming payment files, bank statements, open accounts receivable (AR) ledgers, and remittance advice documents. It executes deterministic, prioritized matching rules to evaluate payment accuracy and assigns explicit record end states.
2. **Capstone 02 (Deduction Identification & Processing Workflow):** Receives target exception records from Stage 1 (specifically Partial Match, UAC, and Query states). It performs variance calculation, tolerance evaluation, deduction code mapping, ledger partial posting, and automated dispute creation.

INTEGRATED FINANCE AI WORKFLOW: CASH APPLICATION & DEDUCTION MANAGEMENT



Capstone 01: Configurable Cash Application Rule Engine

1. Acceptance Criteria

- **External Rule Configuration:** Rules must be configured outside the core execution logic using structured formats (JSON, YAML, or database-driven rulesets) to allow client-specific and deal-specific parameters without code modification.
- **Execution Order & Determinism:** Rules must execute strictly according to assigned priority (Priority 1 to Priority 8). Once a definitive match condition is met, evaluation halts for that record.
- **Deterministic End State Assignment:** The engine must assign one of the six predefined execution end states to every processed bank transaction record.
- **Audit Trail & Observability:** Every execution step must log the evaluated rule ID, matching key criteria, execution timestamp, variance amount, and confidence score for enterprise compliance.

2. Rule Execution End States

Sequence	End State	Description	System Status
1	Open	Bank transaction uploaded but fails to match any Open AR invoice or Remittance record.	OPEN
2	Partial Match	Bank statement line matches Open AR or Remittance advice keys, but presents a non-zero financial variance.	PARTIAL_MATCH
3	Closed	Bank statement	CLOSED

		values fully reconcile against Open AR and Remittance advice records without variance.	
4	UAC (Un-applied Cash)	Payment received with customer identification, but missing invoice reference numbers in Bank Statement and Remittance Advice.	UNAPPLIED_CASH
5	UIC (Un-Identified Cash)	Payment received with no traceable customer, invoice, or remittance context.	UNIDENTIFIED_CASH
6	Query	Record flagged and placed on hold pending external customer or collector feedback.	PENDING_QUERY

3. Priority Rules Engine Specifications

Priority	Bank Statement Fields	ERP AR Open Fields	Remittance Advice Fields	Match Type	Financial Variance Execution Action
1	Customer Name + PO Number + Amount	Customer Name + PO Number (lookup Invoice) + Amount	Not Required	2-Way Match	Unmatched value post to UAC. Set state to PARTIAL_MATCH or

					UAC.
2	Customer Name + Delivery Number + Amount	Customer Name + Delivery Number (lookup Invoice) + Amount	Not Required	2-Way Match	Unmatched value post to UAC. Set state to PARTIAL_MATCH or UAC.
3	Customer Name + Invoice Number + Invoice Date + Amount	Customer Name + Invoice Number + Invoice Date + Amount	Not Required	2-Way Match	Unmatched value post to UAC. Set state to PARTIAL_MATCH or UAC.
4	Customer Name + Invoice Number + Amount	Customer Name + Invoice Number + Amount	Not Required	2-Way Match	Unmatched value post to UAC. Set state to PARTIAL_MATCH or UAC.
5	Customer Bank Payment Details	Customer Name + PO Number + Invoice Number + Invoice Date + Amount	Customer Name + PO Number + Invoice Number + Invoice Date + Amount	3-Way Match	Unmatched value post to UAC. Set state to PARTIAL_MATCH or UAC.
6	Customer Bank Payment Details	Customer Name + PO Number + Invoice Number + Amount	Customer Name + PO Number + Invoice Number + Amount	3-Way Match	Unmatched value post to UAC. Set state to PARTIAL_MATCH or UAC.

7	Customer Name + Invoice Number + Payment Amount	Customer Name + Invoice Number + Invoice Amount	Not Required	2-Way Tolerance Match	If $ \text{Amount}_{\text{Invoice}} - \text{Amount}_{\text{Payment}} \leq \text{Tolerance}_{\text{Configured}}$, auto write-off difference. Set state to CLOSED.
8	Customer Name + Invoice Number + Foreign Currency Amount	Customer Name + Invoice Number + Base Currency Amount	Currency Exchange Rate + Transaction FX Details	3-Way FX Match	Attribute variance to Realized Foreign Exchange Gain/Loss. Set state to CLOSED.

Capstone 02: Deduction Identification & Processing Workflow

When Capstone 01 processes a transaction that results in a payment variance or short payment, execution automatically hands off to Capstone 02.

1. Acceptance Criteria

- **Variance Detection:** Calculate variance using the formula: $\text{Variance} = \text{Amount}_{\text{Invoice}} - \text{Amount}_{\text{Received}}$.
- **Tolerance Application:** If $|\text{Variance}| \leq \text{Tolerance}_{\text{Configured}}$ (e.g., USD 50.00), automatically generate a GL write-off entry and set the record to CLOSED.
- **Reason Extraction & Mapping:** Scan Remittance Advice unstructured text, line-item notes, or structured metadata for dispute indicators and map them to standardized Reason Codes (D01–D07).
- **Ledger Posting Strategy:**
 1. Apply payment against open AR balance equal to $\text{Amount}_{\text{Received}}$.
 2. Mark the received portion as PAID / CLOSED in ERP.
 3. Instantiate a new Dispute Line Item for Variance with the associated Deduction

Reason Code.

- **Dispute Routing:** Route created dispute items to designated department work queues based on reason code classification.

2. Sample Scenario Calculations

Step	Metric	Formula / Value	Resulting System State
Scenario A	Invoice Amount	USD 10,000.00	Input
	Payment Received	USD 9,500.00	Input
	Calculated Variance	$\$10000.00 - 9500.00 = 500.00\$$	Potential Deduction (Exceeds USD 50 Tolerance)
	Workflow Action	Post USD 9,500.00 to AR; Create USD 500.00 Dispute Item.	Hand off to Capstone 02 Dispute Queue.
Scenario B	Invoice Amount	USD 10,000.00	Input
	Payment Received	USD 9,970.00	Input
	Calculated Variance	$\$10000.00 - 9970.00 = 30.00\$$	Within USD 50 Tolerance
	Workflow Action	Post USD 10,000.00 clearing; Write-off USD 30.00 to Auto-Write-Off GL.	Execution End State set to CLOSED.

3. Standard Deduction Reason Mapping & System Codes

Reason Code	Category Description	Operational Cause	Target Resolution Routing Queue
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D01	Pricing Issue	Discrepancy between Purchase Order price and Invoiced unit price.	Sales Operations / Billing
D02	Freight Claim	Dispute regarding freight terms, unauthorized shipping charges, or carrier fees.	Logistics / Supply Chain
D03	Damage Claim	Customer deduction for goods reported damaged during transit or delivery.	Quality Assurance / Warehouse
D04	Tax Difference	Tax calculation mismatch or tax-exempt certificate submission.	Corporate Tax Team
D05	Discount Taken	Cash discount taken outside agreed payment terms.	Credit & Collections
D06	Quantity Shortage	Delivered item count is lower than billed invoice quantity.	Logistics / Warehouse
D07	Promotional Allowance	Customer deducted unapplied volume rebate or marketing allowance.	Trade Marketing / Commercial

End-to-End Technical Execution Flow

